A cutoff principle and conditional closure of the Riemann Hypothesis

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Abstract

This paper states a concrete cutoff principle (a bandlimit/Nyquist hypothesis for prime-frequency observables) and shows that, within the near-field barrier framework, it implies a uniform arithmetic blocker and a height-uniform Carleson bound. Combined with the unconditional far-field zero-freeness of Paper I, this yields a conditional closure of the Riemann Hypothesis. The hypothesis is stated explicitly and kept separate from unconditional analysis.

1 Introduction

The Riemann zeta function $\zeta(s)$ extends meromorphically to $\mathbb{C}$ with a simple pole at $s = 1$. Its nontrivial zeros control fine-scale fluctuations in the distribution of primes, and the Riemann Hypothesis (RH) asserts that every nontrivial zero satisfies $\Re s = \frac{1}{2}$; see [1, 3].

This paper isolates a single explicit additional assumption—a cutoff principle for prime-frequency observables—under which the remaining height-dependent obstruction in the near-field barrier method disappears. The goal is not to claim a new classical theorem, but to present a clean *conditional* implication with sharp claim boundaries.

The obstruction: height growth in the Carleson budget

The near-field barrier mechanism developed in the companion effective paper compares: (i) a *quantized* phase-cost forced by a hypothetical off-critical zero of depth $\eta > 0$, and (ii) an upper bound coming from a Carleson-box energy budget at the corresponding scale $L \sim 2\eta$. In the classical bookkeeping, the budget naturally splits into a prime-layer term (scale-dependent but height-independent) and a zeros term whose dominant growth is proportional to

$$1 \;+\; L \log\langle T\rangle, \qquad \langle T\rangle := \sqrt{1+T^2},$$

where T is the relevant height parameter. This $L \log\langle T\rangle$ growth is the single mechanism that prevents an all-heights exclusion of near-field off-critical zeros within the barrier framework.

A cutoff principle (explicit hypothesis)

We now state the cutoff assumption used in this paper. It is formulated as a bandlimit/Nyquist condition on the test functions used to probe prime-frequency exponential sums.

Hypothesis 1 (Nyquist cutoff for prime-frequency observables)**.** *Fix a parameter $\tau_0 > 0$ (a sampling scale) and set $\Omega_{\max} := \frac{1}{2\tau_0}$. For each local pair (L, t_0) in the regime of interest, let $\Phi_{L,t_0} : \mathbb{R} \to \mathbb{C}$ denote the test function used on the prime side, with Fourier transform*

$$\widehat{\Phi_{L,t_0}}(\xi) \;:=\; \int_{-\infty}^{\infty} \Phi_{L,t_0}(t)\, e^{-it\xi}\, dt.$$

Assume the bandlimit condition

$$\widehat{\Phi_{L,t_0}}(\xi) = 0 \qquad \textit{for all } |\xi| > \Omega_{\max}.$$

Remark 2 (Status of the hypothesis)*.* Hypothesis 1 is *not* a theorem of classical analysis or number theory. In standard analytic frameworks, Fourier support at scale L is expected to grow like $1/L$ when one localizes in physical space. Here we instead postulate a fixed bandwidth ceiling $\Omega_{\max}$ and investigate its consequences for the arithmetic barrier method.

Main conditional implication

The cutoff hypothesis removes the height-growth mechanism in the Carleson budget and converts the effective barrier into a uniform one. The following statement records the near-field implication in the form used in the split project.

Theorem 3 (Nyquist cutoff $\Rightarrow$ uniform near-field exclusion)**.** *Assume Hypothesis 1 and assume the cutoff test family satisfies the uniformity/admissibility conditions invoked in the barrier closure step (Section 4); in particular, assume the uniform bound $M_\Phi < \infty$ of Lemma 5. Then the Riemann zeta function has no zeros in the strip*

$$\Big\{ s \in \mathbb{C} : \; \tfrac{1}{2} < \Re s < 0.6 \Big\}.$$

Corollary 4 (Conditional closure of RH in the three-paper split)**.** *Assume Hypothesis 1 and assume the uniformity/admissibility conditions used to apply Theorem 3 (in particular, $M_\Phi < \infty$ as in Lemma 5). If, in addition, $\zeta(s)$ has no zeros in the half-plane $\{\Re s \geq 0.6\}$ (as proved unconditionally in Paper I of this series), then the Riemann Hypothesis holds.*

Roadmap

Section 2 formalizes the prime-frequency observable and shows that Hypothesis 1 collapses it to a finite prime sum, yielding a *uniform* arithmetic blocker. Section 3 converts this into a height-uniform Carleson-box bound. Finally, Section 4 combines the resulting uniform budget with the classical barrier inequality (proved in Paper II of this series) to obtain Theorem 3 and hence Corollary 4.

2 From Nyquist cutoff to a uniform arithmetic blocker

In the near-field barrier bookkeeping (cf. Paper II), the missing all-heights input can be packaged as a single arithmetic estimate: a scale-uniform bound for a prime exponential sum at the critical weight $p^{-1/2}$. We record the relevant prime-frequency observable and note that Hypothesis 1 forces it to be a *finite* prime sum, yielding a uniform bound by a direct triangle-inequality estimate.

The prime exponential sum

Given (L, t_0), define the *windowed prime exponential sum*

$$S_{L,t_0} \;:=\; \sum_p \frac{\log p}{\sqrt{p}}\, e^{it_0 \log p}\, \widehat{\Phi_{L,t_0}}(\log p). \tag{1}$$

Under Hypothesis 1, the sum in (1) is in fact finite (only primes with $\log p \le \Omega_{\max}$ contribute), so there are no convergence issues. This is the atomic "arithmetic blocker" quantity in the barrier framework of Paper II: bounding $|S_{L,t_0}|$ uniformly in (L, t_0) is the key arithmetic input used to upgrade the near-field exclusion from height-limited to all heights (subject to the admissibility caveat in Remark 7).

Lemma 5 (Nyquist cutoff $\Rightarrow$ uniform arithmetic blocker)**.** *Assume Hypothesis 1. Then S_{L,t_0} can be written as a finite sum over primes $p \le e^{\Omega_{\max}}$, and for every (L, t_0) one has the pointwise bound*

$$|S_{L,t_0}| \;\le\; \Big(\sup_{\xi\in\mathbb{R}} |\widehat{\Phi_{L,t_0}}(\xi)|\Big) \sum_{p\le e^{\Omega_{\max}}} \frac{\log p}{\sqrt{p}}.$$

In particular, if one sets

$$M_\Phi \;:=\; \sup_{(L,t_0)} \sup_{\xi\in\mathbb{R}} |\widehat{\Phi_{L,t_0}}(\xi)|,$$

where the outer supremum is taken over the same parameter set of pairs (L, t_0) for which the test family Φ_{L,t_0} is specified in Hypothesis 1, and if $M_\Phi < \infty$ (for example if $\sup_{(L,t_0)} \|\Phi_{L,t_0}\|_{L^1} < \infty$, since $|\widehat{\Phi}| \le \|\Phi\|_{L^1}$), then for all (L, t_0),

$$|S_{L,t_0}| \;\le\; M_\Phi \sum_{p\le e^{\Omega_{\max}}} \frac{\log p}{\sqrt{p}} \;=:\; K_{\mathrm{Nyq}},$$

so $|S_{L,t_0}|$ is bounded uniformly (independent of L and t_0).

Proof. If $\log p > \Omega_{\max}$ then $\widehat{\Phi_{L,t_0}}(\log p) = 0$ by Hypothesis 1, so only primes $p \le e^{\Omega_{\max}}$ contribute to (1). By the triangle inequality,

$$|S_{L,t_0}| \le \sum_{p\le e^{\Omega_{\max}}} \frac{\log p}{\sqrt{p}}\, |\widehat{\Phi_{L,t_0}}(\log p)| \le \Big(\sup_{\xi\in\mathbb{R}} |\widehat{\Phi_{L,t_0}}(\xi)|\Big) \sum_{p\le e^{\Omega_{\max}}} \frac{\log p}{\sqrt{p}},$$

which is finite and depends only on $\Omega_{\max}$ and the pointwise bound $\sup_\xi |\widehat{\Phi_{L,t_0}}(\xi)|$. If $M_\Phi < \infty$ then the stated uniform bound follows immediately. □

Remark 6 (Why this is the point of the cutoff)*.* Without a fixed bandwidth ceiling, localization at scale L typically forces Fourier support to expand like $1/L$. For instance, if $\Phi_{L,t_0}(t) = L^{-1}\phi((t - t_0)/L)$ then $\widehat{\Phi_{L,t_0}}(\xi) = e^{-it_0\xi}\widehat{\phi}(L\xi)$, so any fixed-frequency support for $\widehat{\phi}$ becomes support of size $\asymp 1/L$ for $\widehat{\Phi_{L,t_0}}$ as $L \downarrow 0$. In that regime, the prime sum (1) naturally involves primes with $\log p$ as large as a constant multiple of $1/L$, and trivial bounds grow rapidly as $L \downarrow 0$. Hypothesis 1 rules out this growth mechanism by forcing (1) to be a finite sum over primes $p \le e^{\Omega_{\max}}$ for *every* scale.

Remark 7 (Two blockers to an unconditional upgrade)*.* It is important to separate what Lemma 5 does from what it does not do.

- **Admissibility.** One must check that the barrier argument can be run using only test functions satisfying Hypothesis 1 for all relevant (L, t_0).
- **Bridge gap.** The hypothesis itself is not presently implied by classical number theory; replacing it by a proved arithmetic estimate is a main remaining hard step.

3 From the arithmetic blocker to a uniform Carleson bound

The quantity S_{L,t_0} in (1) is the "atomic arithmetic blocker" in the sense that it captures the prime-frequency input that must be controlled uniformly in the near-field barrier bookkeeping (cf. Paper II of this series). This section records a clean analytic consequence of the cutoff hypothesis for the associated *prime-frequency field*: under Hypothesis 1, this field is a finite superposition of harmonic modes and its Carleson-box energy is bounded uniformly in both scale L and center t_0.

Carleson boxes and box energy

Work in half-plane coordinates $s = \frac{1}{2} + \sigma + it$ with $\sigma > 0$. Given $L > 0$ and $t_0 \in \mathbb{R}$, set $I := I_{L,t_0} = [t_0 - L, t_0 + L]$ and let $Q(\alpha I) := I \times (0, \alpha|I|]$ be the Carleson box of fixed aperture $\alpha > 1$. For a harmonic function U on $Q(\alpha I)$ define its (local) box-energy ratio by

$$\mathcal{C}_{\mathrm{box}}(U; I) \;:=\; \frac{1}{|I|} \int_{Q(\alpha I)} |\nabla U(\sigma, t)|^2 \, \sigma \, d\sigma \, dt. \tag{2}$$

In the near-field barrier framework, U is the (real-valued) harmonic log-modulus potential attached to a normalized zeta-ratio (after local neutralization), and bounding (2) uniformly is the "budget" input used to close the barrier for all heights. Here we record a clean uniform bound for the explicit cutoff prime field defined below; the admissibility discussion in Remark 7 addresses how this surrogate input is used in the closure step.

Finite prime-frequency support under the cutoff

Under Hypothesis 1, only primes with $\log p \le \Omega_{\max}$ contribute to (1). At the level of the prime-frequency observable, this replaces an effectively unbounded-frequency superposition (as $L \downarrow 0$) by a *finite* one. To make the resulting uniformity explicit, define the complex prime-frequency field on $Q(\alpha I)$ by

$$W_{L,t_0}(\sigma, t) \;:=\; \sum_{p \le e^{\Omega_{\max}}} \frac{\log p}{\sqrt{p}} \, \widehat{\Phi_{L,t_0}}(\log p) \, e^{-(\sigma - it)\log p}. \tag{3}$$

Each summand is harmonic in (σ, t), hence $\Re W_{L,t_0}$ is harmonic and real-valued. With the Fourier convention in Hypothesis 1, one has the exact identity

$$W_{L,t_0}(0, t_0) \;=\; S_{L,t_0}.$$

Corollary 8 (Uniform Carleson bound for the cutoff prime field)**.** *Assume Hypothesis 1 and suppose the test family satisfies a uniform bound*

$$M_\Phi \;:=\; \sup_{(L,t_0)} \; \sup_{\xi \in \mathbb{R}} |\widehat{\Phi_{L,t_0}}(\xi)| \;<\; \infty.$$

Here the outer supremum is taken over the same parameter range of pairs (L, t_0) for which the family Φ_{L,t_0} is specified in Hypothesis 1. Then there exists a finite constant $C_{\mathrm{Nyq}} < \infty$ (depending only on α, $\Omega_{\max}$, and M_Φ) such that for every (L, t_0),

$$\mathcal{C}_{\mathrm{box}}(\Re W_{L,t_0}; \, I_{L,t_0}) \;\le\; C_{\mathrm{Nyq}}.$$

In particular, the Carleson-box energy budget is uniform in L and t_0, and hence does not *grow with height.*

Proof. Write $\omega_p := \log p \le \Omega_{\max}$ and $c_p := \frac{\log p}{\sqrt{p}}\,\widehat{\Phi_{L,t_0}}(\log p)$. For the basic mode $g_p(\sigma,t) := e^{-(\sigma-it)\omega_p}$ we have $|\partial_\sigma g_p| = |\partial_t g_p| = \omega_p e^{-\sigma\omega_p}$, hence $|\nabla g_p|^2 \le 2\omega_p^2 e^{-2\sigma\omega_p}$. Since $W_{L,t_0} = \sum c_p g_p$ is a finite sum, we may use $|\sum v_p|^2 \le N \sum |v_p|^2$ with $N := \#\{p \le e^{\Omega_{\max}}\}$ to obtain

$$|\nabla W_{L,t_0}|^2 \le N \sum_{p \le e^{\Omega_{\max}}} |c_p|^2\, |\nabla g_p|^2 \le 2N \sum_{p \le e^{\Omega_{\max}}} |c_p|^2\, \omega_p^2 e^{-2\sigma\omega_p}.$$

Integrating over $Q(\alpha I)$ and using

$$\int_0^{\alpha|I|} \sigma\, e^{-2\sigma\omega_p}\, d\sigma \;\le\; \int_0^{\infty} \sigma\, e^{-2\sigma\omega_p}\, d\sigma \;=\; \frac{1}{4\omega_p^2}$$

yields

$$\int_{Q(\alpha I)} |\nabla W_{L,t_0}|^2\, \sigma\, d\sigma\, dt \le 2N|I| \sum_{p \le e^{\Omega_{\max}}} |c_p|^2\, \omega_p^2 \cdot \frac{1}{4\omega_p^2} = \frac{N}{2}\, |I| \sum_{p \le e^{\Omega_{\max}}} |c_p|^2.$$

Since $|c_p| \le M_\Phi \frac{\log p}{\sqrt{p}}$, we obtain

$$\frac{1}{|I|} \int_{Q(\alpha I)} |\nabla W_{L,t_0}|^2\, \sigma\, d\sigma\, dt \le \frac{N}{2}\, M_\Phi^2 \sum_{p \le e^{\Omega_{\max}}} \frac{(\log p)^2}{p} =: \; C_{\mathrm{Nyq}} < \infty.$$

Finally, since $|\nabla(\Re W_{L,t_0})| \le |\nabla W_{L,t_0}|$ pointwise, the same bound holds with W_{L,t_0} replaced by $\Re W_{L,t_0}$. □

Remark 9 (How this closes the height dependence). In the effective bookkeeping of Paper II, the Carleson budget is bounded by a prime-layer contribution together with a height-dependent zero-layer term of size $1 + L\log\langle T\rangle$ coming from the density of critical-line zeros near height T. The point of Hypothesis 1 is to enable a different all-scales input: at every scale, the prime-frequency field becomes a finite superposition of modes, so its box energy is bounded by a uniform constant as in Corollary 8. Subject to the admissibility discussion in Remark 7, this provides the height-uniform budget input used in the barrier closure step (Section 4).

4 Barrier closure and conditional RH

This section records the final implication chain. The two analytic ingredients behind the near-field barrier are classical:

- a *quantized* lower bound on localized boundary phase mass forced by an off-critical zero (a Blaschke/Poisson "trigger");
- a CR–Green inequality that upper-bounds the same localized phase mass by the square root of a Carleson-box energy budget.

These are proved in Paper II of this series. The additional (conditional) input isolated here is the cutoff-based Carleson estimate of Corollary 8, which is uniform in height for the cutoff prime-frequency field; cf. the admissibility caveat in Remark 7. Note that Corollary 8 also assumes the uniform test-family bound $M_\Phi < \infty$, and we will invoke that hypothesis here as well.

Barrier inequality (cost versus budget)

Fix a depth parameter $\eta \in (0, 0.1)$ and set the associated near-field scale $L := 2\eta$. Let $\rho = \frac{1}{2} + \eta + i\gamma$ be a hypothetical off-critical zero of $\zeta(s)$. We fix a smooth flat-top window as in Paper II: choose an even $\psi : \mathbb{R} \to [0, 1]$ with $\psi \equiv 1$ on $[-1, 1]$ and $\operatorname{supp} \psi \subset [-2, 2]$, and define

$$\psi_{L,t_0}(t) \;:=\; \psi\Big(\frac{t - t_0}{L}\Big).$$

As in the barrier setup of Paper II, let w be a boundary phase (defined modulo 2π) such that $-w'$ is a nonnegative distribution and, in the sense of distributions,

$$-w'(t) \;\geq\; \frac{2\eta}{(t-\gamma)^2 + \eta^2}.$$

Let U denote the harmonic log-modulus potential attached (after local neutralization, as in Paper II) to a normalized zeta-ratio on the Carleson box above the support interval

$$I^* \;:=\; [\gamma - 2L, \gamma + 2L].$$

Proposition 10 (Energy barrier inequality)**.** *With the notation above, there exists a finite constant $C(\psi) > 0$ (depending only on the fixed window profile and box aperture) such that*

$$\int_{\mathbb{R}} \psi_{L,\gamma}(t)\,(-w'(t))\,dt \;\geq\; L_{\mathrm{rec}} \qquad \textit{and} \qquad \int_{\mathbb{R}} \psi_{L,\gamma}(t)\,(-w'(t))\,dt \;\leq\; C(\psi)\sqrt{4L\,\mathcal{C}_{\mathrm{box}}(U; I^*)}. \tag{4}$$

Equivalently,

$$L \cdot \mathcal{C}_{\mathrm{box}}(U; I^*) \;\geq\; \frac{L_{\mathrm{rec}}^2}{4\,C(\psi)^2}, \qquad L_{\mathrm{rec}} := 4\arctan(2).$$

Proof. This is exactly the barrier inequality proved in the companion effective paper of this series (Paper II), specialized to the present notation. For completeness we summarize the two inputs.

Lower bound (Blaschke trigger). The distribution $-w'$ is nonnegative and satisfies the explicit lower bound $-w'(t) \geq \frac{2\eta}{(t-\gamma)^2+\eta^2}$ in the sense of distributions. Testing against the flat-top window $\psi_{L,\gamma}$ with $L = 2\eta$ yields the universal constant $L_{\mathrm{rec}} = 4\arctan(2)$.

Upper bound (CR–Green). After local neutralization, one writes $\log F = U + iV$ on the relevant Carleson box and uses the Cauchy–Riemann identity $\partial_t V = \partial_\sigma U$ (in distributions) together with Green's identity to express the boundary pairing against $\psi_{L,\gamma}$ as a Dirichlet pairing. Cauchy–Schwarz then gives the bound in terms of the box energy $\mathcal{C}_{\mathrm{box}}(U; I^*)$, with the constant $C(\psi)$ depending only on the fixed aperture and window profile. □

Uniform closure of the near-field strip (conditional)

In the effective bookkeeping of Paper II, the Carleson budget is bounded by a prime-layer contribution together with a height-dependent zero-layer term of size $\asymp 1 + L\log\langle|\gamma|\rangle$ coming from the density of critical-line zeros near height $|\gamma|$. The purpose of Hypothesis 1 is to supply a height-uniform substitute for the prime-frequency input: at every scale, the cutoff prime field is a finite superposition of modes, and Corollary 8 gives a bound for its box-energy ratio that is uniform in γ. Subject to the admissibility discussion in Remark 7 (that the barrier argument may be run within a cutoff-admissible test family), this provides the height-uniform budget input used in the closure step.

Proof of Theorem 3 (conditional). Assume Hypothesis 1 and the uniform bound $M_\Phi < \infty$ so that Corollary 8 applies. If there were a zero $\rho = \frac{1}{2} + \eta + i\gamma$ with $0 < \eta < 0.1$, then Proposition 10 would force a fixed positive lower bound on the localized phase mass while simultaneously upper-bounding it by a Carleson budget at the same scale $L = 2\eta$. In Paper II, the only appearance of the height parameter is through the budget bound; the obstruction is a zero-layer term producing the growth $\asymp L \log\langle|\gamma|\rangle$. Under Hypothesis 1, the cutoff prime field $W_{L,\gamma}$ of Section 3 is a finite superposition of modes and satisfies the height-uniform Carleson estimate of Corollary 8. Assuming admissibility as in Remark 7, this removes the height-growth escape in the barrier closure and excludes such a zero for all γ. Therefore the strip $\{\frac{1}{2} < \Re s < 0.6\}$ is zero-free. □

Conditional RH closure in the three-paper split

We now close the implication to RH using the unconditional far-field paper.

Proof of Corollary 4. Assume Hypothesis 1 and the uniform bound $M_\Phi < \infty$ so that Theorem 3 applies. By Theorem 3, $\zeta(s)$ has no zeros in the strip $\{\frac{1}{2} < \Re s < 0.6\}$. By the unconditional far-field result of Paper I, $\zeta(s)$ has no zeros in the half-plane $\{\Re s \geq 0.6\}$. Therefore the completed zeta function $\xi(s)$ has no zeros with $\Re s > \frac{1}{2}$. By the functional equation $\xi(s) = \xi(1-s)$, the zeros of ξ are symmetric with respect to $\Re s = \frac{1}{2}$, hence ξ has no zeros with $\Re s < \frac{1}{2}$ either. Thus every zero of ξ lies on $\Re s = \frac{1}{2}$, i.e. RH holds conditionally. □

Conclusion and limitations (conditional status)

This paper isolates a single explicit additional hypothesis—the Nyquist cutoff hypothesis (Hypothesis 1)—and records its consequences within the near-field barrier framework. Under this hypothesis, the prime-frequency observable collapses to a finite sum (Lemma 5). Assuming an admissible cutoff test family satisfying the uniform bound required in Corollary 8 (in particular $M_\Phi < \infty$), one obtains the height-uniform Carleson estimate used in the barrier closure step and hence the conditional elimination of near-field off-critical zeros (Theorem 3). Combined with the unconditional far-field zero-freeness proved in Paper I, this yields conditional RH (Corollary 4).

What is proved (conditional). The conclusions in Theorem 3 and Corollary 4 are **conditional** on Hypothesis 1, together with the explicit admissibility/uniformity assumptions needed to apply the cutoff Carleson estimate (Corollary 8) in the barrier closure step (Section 4); in particular the uniform bound $M_\Phi < \infty$. No claim is made here that these assumptions follow from classical number theory.

Where the hypothesis enters. The near-field barrier inequality is classical and the vortex cost is height-independent. In the classical bookkeeping (Paper II), height enters only through the budget term, which grows like $L \log\langle T\rangle$. Hypothesis 1 enters by enforcing a fixed bandwidth ceiling on the prime-side test family, removing the high-frequency growth mechanism behind this term within the cutoff-based input.

Normalization caveat. The uniform bounds in Lemma 5 and Corollary 8 depend on an explicit uniform control of the test family, for example a bound $M_\Phi < \infty$ defined by

$$M_\Phi \;:=\; \sup_{(L,t_0)} \sup_{\xi\in\mathbb{R}} |\widehat{\Phi_{L,t_0}}(\xi)|.$$

Any implementation of the cutoff principle must specify admissible test functions and verify the barrier setup remains valid within that admissible class.

Unconditional content. This paper contains *no* new unconditional zero-free statement beyond what is proved in Paper I (far-field) and Paper II (effective near-field). It is included solely to record the clean logical implication: *if* a fixed Nyquist cutoff for prime-frequency observables holds (within an admissible test family giving the stated uniform bounds), *then* the near-field barrier admits a height-uniform closure and RH follows (conditionally).

Statements and Declarations

Competing interests. The author declares no competing interests.

Data and materials availability. This paper introduces no new computational artifacts beyond those already present in the repository. All files needed to compile this manuscript are included in the repository. The unconditional far-field certification artifacts and verifier are documented in `README.md` and in Paper I of this series (`paper1_farfield.tex`).

Reproducibility. The conditional implications in this paper are reproducible from the stated hypotheses (including Hypothesis 1 together with the admissibility/uniformity assumptions invoked in Section 4) and the cited analytic inputs. In particular, the barrier inequality and its effective bookkeeping are proved in Paper II (`paper2_energy_barrier.tex`), while the certified far-field zero-freeness used in Corollary 4 is proved in Paper I (`paper1_farfield.tex`).

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